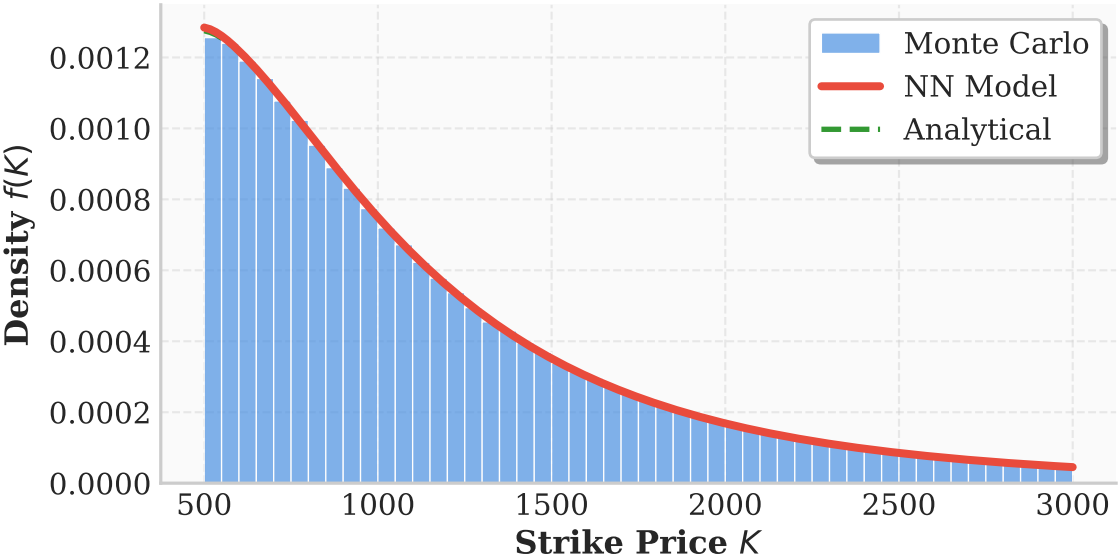
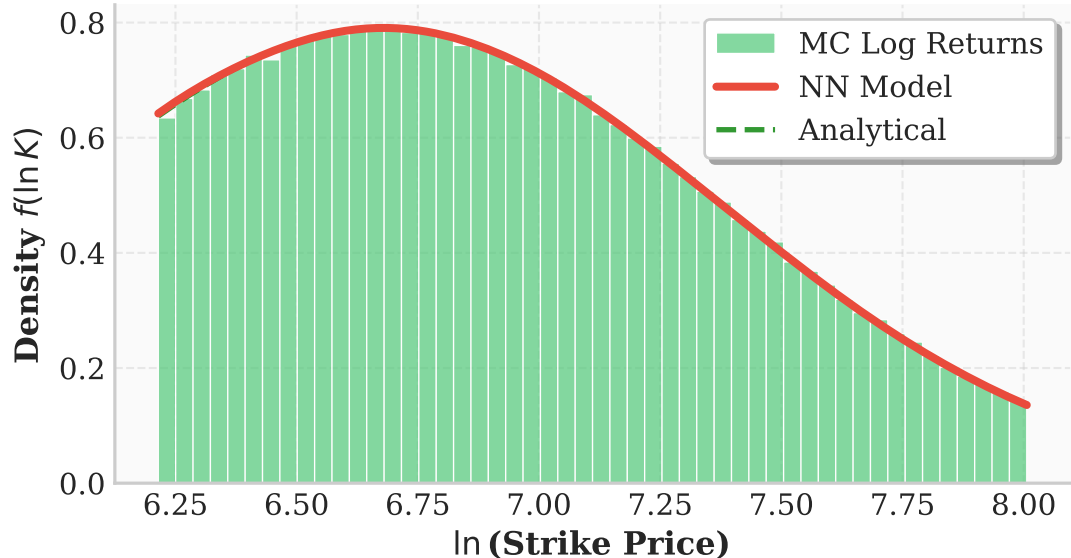


PDF Analysis: Neural Network Synthetic Local Volatility Model
Monte Carlo with $dS_t = rS_tdt + \sigma_{NN}(t, S)S_tdW_t$

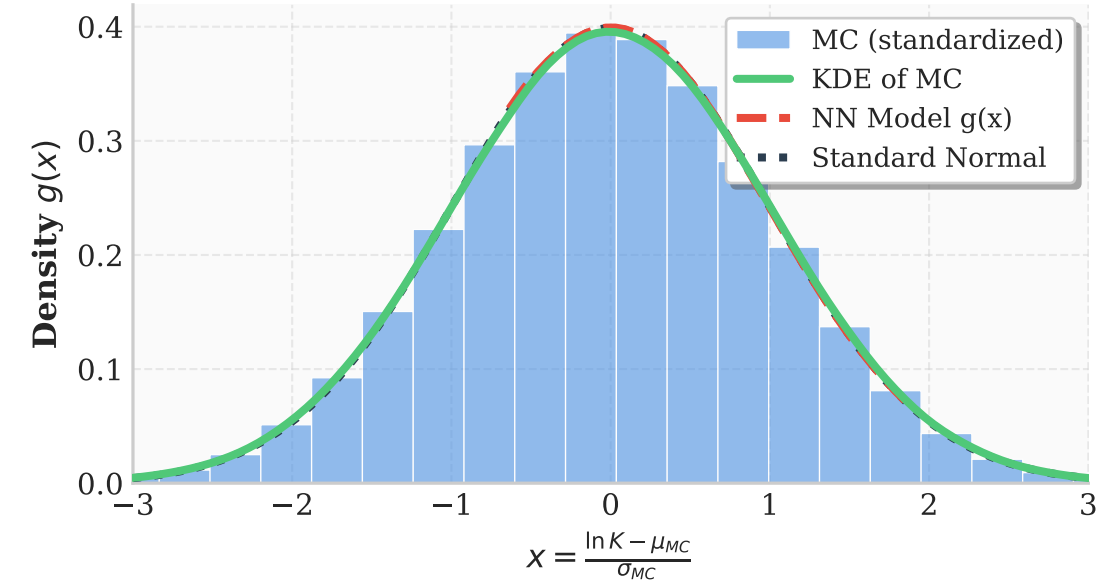
Strike Distribution (T = 0.50)



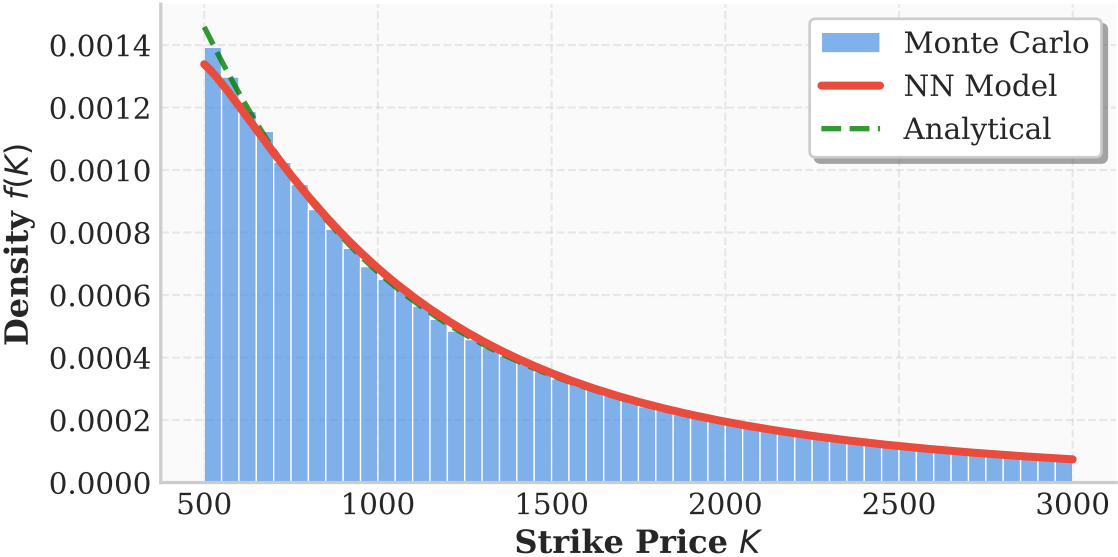
Log-Normal Distribution (T = 0.50)



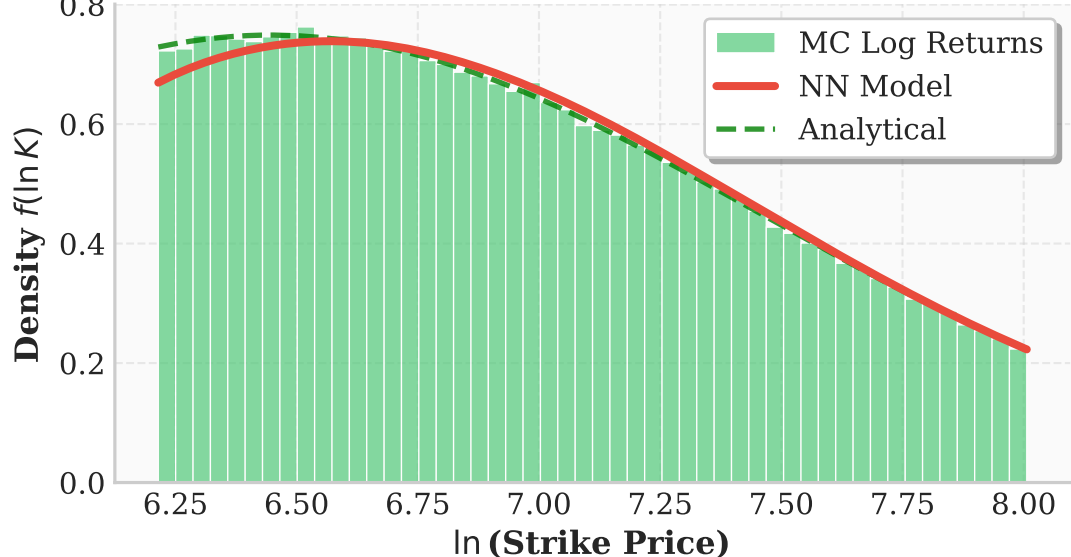
Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.68, \sigma_{MC}=0.71 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\bar{\mu}_{model}=6.93, \sigma_{model}=0.46$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.01$



Strike Distribution (T = 1.00)



Log-Normal Distribution (T = 1.00)



Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.45, \sigma_{MC}=1.00 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\bar{\mu}_{model}=6.96, \sigma_{model}=0.48$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.00, \text{ExKurt}=0.00$

